

Karnataka Storage Decision Brief

Executive Summary — one page

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What this brief examines

On 4 June 2026 the Karnataka Electricity Regulatory Commission ("the Commission") notified the **draft Karnataka Electricity Regulatory Commission (Connectivity and General Network Access to the Intra-State Transmission and State Distribution System) Regulations, 2026** — Notification No. KERC/F-32/V-27/2083, published in the Karnataka Gazette on 11 June 2026. The draft proposes commencement from **1 October 2026**.

It can materially alter the economics of open access and captive solar projects in Karnataka in three places. **How far it reaches a given project depends on which banking cohort that project falls in** — see below.

All regulation references below are to the draft. The final notification is awaited. Nothing in this brief is an operative entitlement today.

Three changes, and the date attaching to each

1 · January 2027 — a ninety-day window

Under **Regulation 5.10(b)** (transmission) and **5.20(b)** (distribution), a solar-based Renewable Energy Generating Station ("REGS") or a hybrid station with a solar component, which holds in-principle or final grant of Connectivity or whose GNA is effective, is **converted into an entity with solar hour access** within one week after the expiry of three months from the date these Regulations take effect.

The consequence is that **injection scheduling rights during non-solar hours are lost**, save to the extent of capacity from a non-solar source.

Regulation 5.10(b) preserves a right to apply for additional capacity under Regulation 5.2 or 5.10(a) **within that same three-month period**. There is no provision in the draft for extension of that window.

2 · Already in force — the slot restriction on banking

This does not depend on the draft. Clause 12.e of the KERC (Terms and Conditions for Open Access) Regulations, 2025, notified 26 March 2025, already provides the same mechanism that appears at draft Regulation 39.3:

Energy banked in	May be drawn in
ToD Slot-1 and Slot-3	Slots 1, 2, 3 and 4
ToD Slot-2 and Slot-4	Slot-2 and Slot-4 only

Solar generation falls almost entirely within Slot-2 (09:00–18:00). Slot-3 (18:00–22:00) carries a ToD premium of +100 paise for eight months of the year.

Banked solar energy therefore cannot be drawn during the evening peak, at any price.

Unutilised banked energy lapses at the end of each month (draft Regulation 39.2; Clause 12.e today), with Renewable Energy Certificate ("REC") eligibility for the lapsed quantum.

3 · 31 March 2030 — and, for many, an earlier date

Draft **Regulation 39.1** fixes a single terminal date for monthly banking. The operative regulation today confers the entitlement for **five years from the effective date of each agreement** — a rolling right. The draft replaces it with one common date, so the later the Wheeling and Banking Agreement ("WBA"), the shorter the entitlement.

The proviso to Regulation 39.1 goes further: **WBAs executed before 13 January 2023 (short-term open access) and before 2 January 2023 (long-term open access) are not entitled to the banking facility from the date of renewal.**

On the Karnataka SLDC Wheeling and Banking list dated 9 July 2024 (556 entries / 4,924.08 MW), that cohort

comprises **358 agreements / 3,725 MW — 76 per cent of the W&B fleet**. The concentration peaks in **FY 2027-28: 61 agreements / 1,024 MW**.

The conclusion is not "procure a battery"

This is the central finding of the brief, and it runs against the prevailing advice.

For an HT-2(a) industrial consumer, per kWh of incremental solar generation:

Route	Value
Banking at 8 per cent in kind — where the consumer has night-time load	₹5.54
Physical storage — discharged into Slot-3	₹3.93
Banking, where there is no Slot-2 / Slot-4 offtake	₹0 (<i>month-end lapse</i>)

This is computed on the **most storage-favourable** capacity charge traced for 2025 — ₹1.48 lakh per MW per month (APTRANSCO tender, December 2025). At benchmark rates the gap widens from ₹1.61 to ₹2.81 per kWh.

Banking is finished for the evening peak. For night-time offtake it remains worth ₹1.61/kWh more than storage.

Three break-evens, none of them near:

Variable	Present	Break-even
Banking charge	8 per cent	34.7 per cent
Levelised storage cost	₹2.56/kWh discharged	₹0.67/kWh
Slot-3 ToD adder	₹1.00/kWh	₹4.24/kWh

What the decision actually turns on — two questions, in this order

Question 1 · Which banking cohort does the project fall in?

Karnataka does not operate one banking regime. It operates two, in parallel.

	Legacy cohort	Current cohort
Who	Non-REC projects that applied between 1 April 2018 and the commencement of the GEOA Regulations, 2022	Subsequent WBAs and renewals
Settlement period · charge	Annual · 2 per cent	Monthly · 8 per cent
Month-end lapse · slot restriction	No	Yes
Status	Clause 6 preserves this entitlement " subject to the outcome before the Hon'ble Supreme Court " (<i>the regulation's own words; the proceeding has not been independently verified here</i>)	In force

Source: OA Regulations, 2025, Clause 6 and Clause 12.e.

Every figure above is computed for the current cohort. Applying them to a legacy-cohort project would be wrong.

And the draft is **silent on the legacy cohort**. Regulation 39 does not mention it. Whether the 8 per cent charge, the month-end lapse and the 31 March 2030 sunset apply to it, and under what saving or transition provision, is not stated.

Question 2 · How many shifts does the consumer run?

Shift pattern	Slot-3 load	Slot-4 load	Outcome
Three shifts	Yes	Yes	Banking
Two shifts	Yes	No	Storage — this is the real customer
Single shift	No	No	Neither
Day plus night	No	Yes	Banking

Do not ask for utilisation. Ask for the shift pattern. It is obtained in one sentence, and it identifies the likely case — though twelve months of 15-minute data is what settles it.

A consumer on **DERS** — the incremental-consumption tariff under Clause 6.14 of the Tariff Order dated 27 March 2025, a flat ₹5/kWh above base with ToD not applicable — has no evening premium at all. The gap widens to ₹2.53/kWh.

On this model, therefore, DERS weakens the case for storage rather than strengthening it — and DERS eligibility should be established before any storage economics are run.

Three provisions in the draft that are not being discussed

1 · A third party's battery can fit inside your existing Connectivity. Regulation 5.2 (transmission) and 5.12 (distribution) permit a generating station or ESS, with the prior approval of the Nodal Agency, to add generation capacity or an Energy Storage System **within the sanctioned quantum of Connectivity — including capacity owned by another entity.** No fresh application, no additional terminal bay, no new dedicated line. *Subject to Nodal Agency approval, available headroom in the sanctioned quantum, and project-specific conditions:* the existing Connectivity grantee remains responsible for Grid Code compliance as Lead Generator or Lead ESS, and therefore for deviation exposure arising from another party's asset. **Eight contractual questions have to be closed before this is bankable — Section 2.2.**

2 · Night-hour access on someone else's terminal bay. Regulation 5.10(a)(ii) permits a wind-based REGS or an ESS to seek Connectivity with non-solar hour access at a terminal bay "*which is already allocated to another REGS or Renewable Power Park, with solar hour access.*" A solar project's evacuation infrastructure stands idle at night. Under the proviso to Regulation 8.2(a), Conn-BG2 is not required where the bay was constructed by another Connectivity grantee **at his own cost.**

3 · One sentence that may save a substantial sum annually. Regulation 31.3 escalates transmission and wheeling charges by 1.25×, 1.5× and 2.0× from the fourth Temporary GNA ("T-GNA") transaction in a financial year — but expressly "*irrespective of period of T-GNA sought.*" Regulation 27.1 permits a single T-GNA application for any period **from one time block up to eleven months.** On that reading: **potentially one transaction, one ₹5,000 fee, no multiplier — subject to the SLDC procedure under Regulation 49.2 defining "transaction",** which the draft does not.

What this brief is not

It is not legal advice, and not investment advice. It does not use project-specific 15-minute interval data, which is not publicly available and which makes the largest difference to the outcome.

All regulation numbers are drawn from the **draft**; the final notification is pending. Method, assumptions, the source register and eleven recorded limitations are set out in the Annexure.

In one sentence

A battery is early for most consumers today. But every project has a date, and it falls either on the renewal of its WBA or on 31 March 2030.

Find that date, and decide eighteen months ahead of it — because all three defensive routes in the draft depend on things that queue.

This brief is free and it may be wrong. If any figure or conclusion appears incorrect, please say where it fails — corrections will be recorded, with attribution, in the next version.